Government College of Engineering, Jalgaon
(An Autonomous Institute of Government of Maharashtra)

Name :	**PRN :**
Class : L. Y. B.Tech Computer	**Batch:**
Subject : CO456U Cloud Computing Lab	**Sem :** VIIIth
Course Teacher: Prof. S. D. Cheke	**Academic Year :** 2025-26
Date of Performance :	**Date of Completion :**

Practical No - 1

Aim: Launch EC2 Ubuntu and demonstrate connecting it using mobaxterm

Theory :

Objective: To launch a virtual server (Ubuntu instance) on AWS EC2 and establish a secure remote connection using MobaXterm via SSH.

What is AWS EC2?

Amazon EC2 **(Elastic Compute Cloud)** is a cloud computing service provided by Amazon Web Services that allows users to create and manage virtual machines (instances) over the internet.

- Provides scalable computing capacity
- Supports multiple OS (Linux, Windows)
- Pay-as-you-go pricing model

What is Ubuntu?

Ubuntu is a popular open-source Linux operating system widely used for servers due to its stability, security, and ease of use.

What is SSH?

SSH is a secure protocol used to remotely access and manage servers over a network.

- Uses encryption for secure communication
- Requires authentication (key pair or password)

What is MobaXterm?

MobaXterm is a terminal application for Windows that supports SSH, file transfer (SFTP), and remote computing.

- Easy GUI-based SSH connection
- Built-in file explorer
- Supports key-based authentication

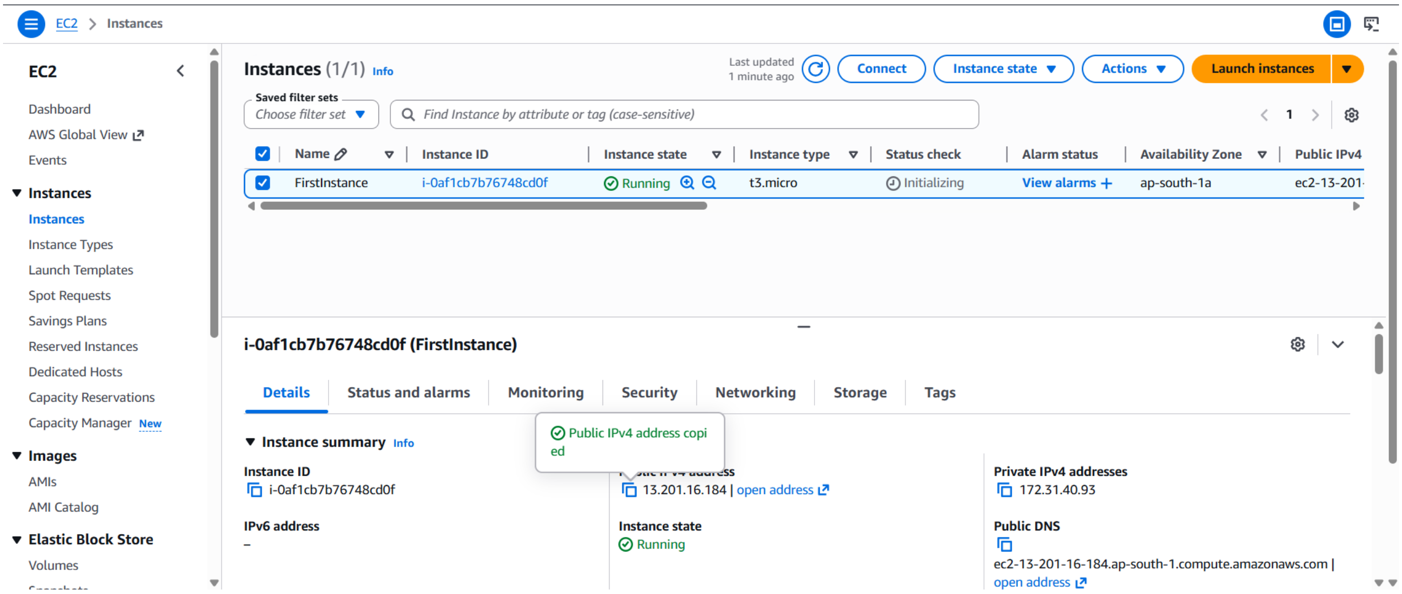
Requirements

- AWS Account
- Internet connection
- MobaXterm installed
- Key pair (.pem file)

Procedure / Steps

Step 1: Login to AWS Console

1. Go to AWS Management Console
2. Login using your credentials
3. Navigate to EC2 Dashboard



Step 2: Launch EC2 Instance

1. Click **"Launch Instance"**
2. Configure the following:
    1. **Name**: Ubuntu-Server
    2. **AMI (OS)**: Select Ubuntu Server
    3. **Instance Type**: t2.micro (Free tier)
    4. **Key Pair**: Create new key pair
        1. Name: my-key
        2. Type: RSA
        3. Format: .pem
    5. **Network Settings**:

        1. Allow SSH (Port 22)
3. Click **Launch Instance**

Step 3: Check Instance Status

- Wait until instance state = **Running**
- Copy the **Public IPv4 address**

Step 4: Configure Security Group

Ensure:

- SSH (Port 22) is allowed
- Source: Anywhere (0.0.0.0/0) or your IP

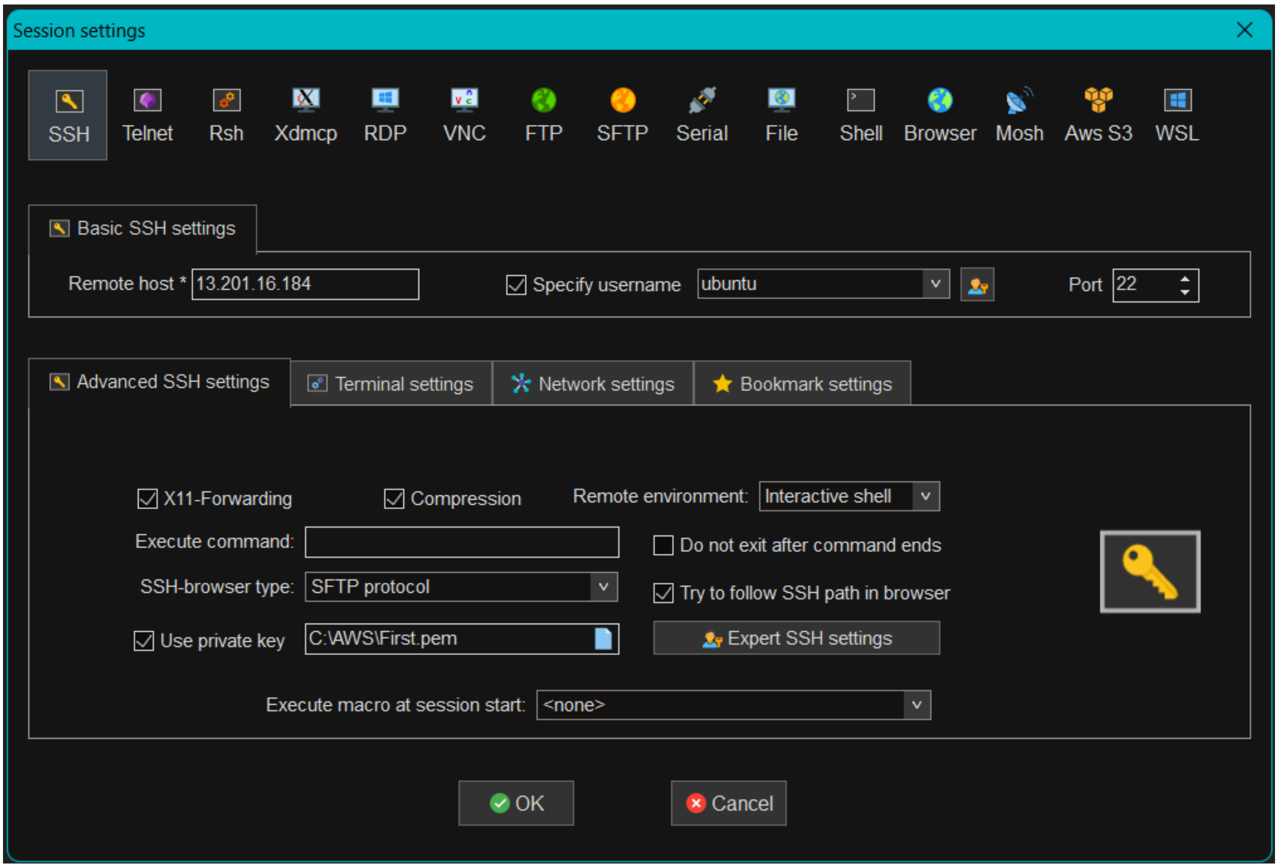
Step 5: Open MobaXterm

1. Launch **MobaXterm**
2. Click **Session → SSH**

Step 6: Configure SSH Connection

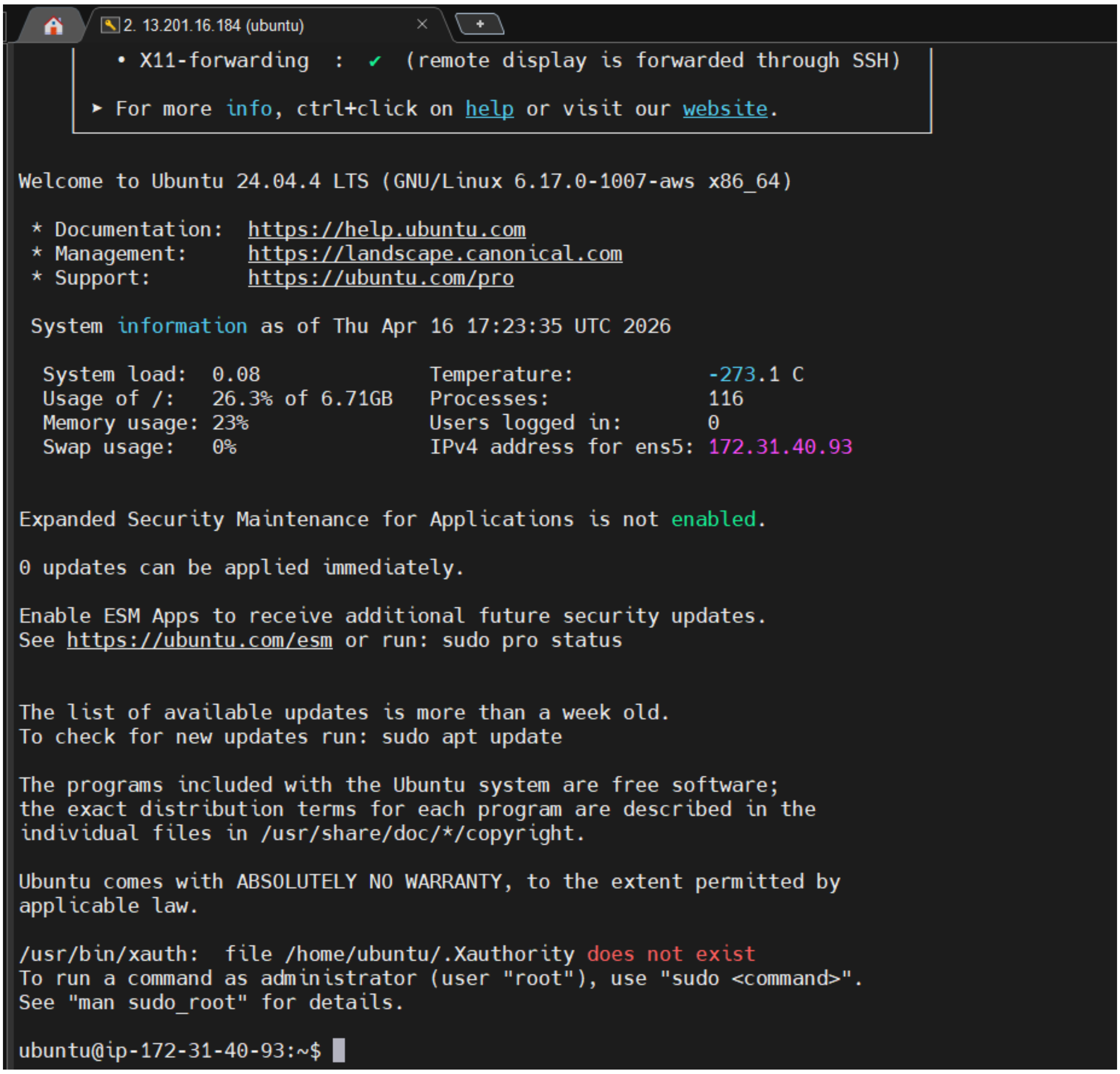
Enter:

- **Remote Host**: Public IP of EC2
- **Specify Username**: ubuntu
- Go to **Advanced SSH settings**
- Select your .pem key file



Step 7: Connect to EC2

1. Click **OK**
2. Accept security prompt
3. Terminal will open → Connected to Ubuntu server



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2. 13.201.16.184 (ubuntu)
    • X11-forwarding  :  ✔  (remote display is forwarded through SSH)

  ➤ For more info, ctrl+click on help or visit our website.

Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Apr 16 17:23:35 UTC 2026

  System load:  0.08              Temperature:           -273.1 C
  Usage of /:   26.3% of 6.71GB   Processes:             116
  Memory usage: 23%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.40.93


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

/usr/bin/xauth:  file /home/ubuntu/.Xauthority does not exist
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-40-93:~$
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Successfully launched an Ubuntu EC2 instance on AWS and established a secure SSH connection using MobaXterm.

Conclusion: In this way, we have successfully implemented and completed the process of launching an EC2 Ubuntu instance and connecting it using MobaXterm.

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